A Novel Graph-Based Motion Planner of Multi-Mobile Robot Systems With Formation and Obstacle Constraints

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***Abstract*—Multi-mobile robot systems (MMRSs) show great advantages over one single robot in many applications. However, the robots are required to form desired task-specified formations, making feasible motions decrease significantly. Thus, it is challenging to determine whether the robots can pass through an obstructed environment under formation constraints, especially in an obstacle-rich environment. Furthermore, is there an optimal path for the robots? To deal with the two problems, a novel graph-based motion planner is proposed in this article. Valid configurations of the system are defined to satisfy both formation and obstacle constraints. Then, the whole valid configuration space is identified and mapped to an undirected graph. The breadth-first search (BFS) method is employed on the graph to answer the question of whether there is a feasible path on the graph. Finally, an optimal path will be planned on the updated graph, considering the cost of path length and formation preference. Simulation results show that the planner can be applied to get optimal motions of robots under formation constraints in obstacle-rich environments. In addition, different types of constraints are considered to verify the generality.**

***Index Terms*—Formation constraints, motion planning, multi-mobile robot systems (MMRS), obstacle-rich environment.**

I. Introduction

In recent years, multi-mobile robot systems (MMRS) have attracted increasing attention from robotics researchers. Compared with a single robot, the robots can cooperate with each other to achieve better system robustness and flexibility. Through collaboration among robots, complex tasks can be accomplished, such as search and rescue [1], [2], exploring [3], assembling [4], and transporting [5]. Moreover, MMRS can replace the bigger single robot in scenarios with a restricted and cramped environment, where there is not enough space. In many applications of MMRS mentioned above, the robots are usually required to form desired formations to achieve cooperation, such as the object will connect the robots to form a whole system during cooperative transporting. Accordingly, the flexibility of each robot will be decreased, since it has to satisfy formation constraints. In the meanwhile, the robots also need to avoid obstacles when working, which requires them to be as flexible as possible. Conflicts will arise if simultaneously considering formation constraints and obstacle avoidance when planning motions of robots, especially in a dense environment with multiple obstacles. Therefore, it is a challenging problem to determine whether MMRS can pass through an obstructed environment under task-specified formation constraints. Moreover, if it is possible, what is the optimal path for the robots to fulfill the task?

Thus, we focus on multimobile robot motion planning problems under formation constraints in obstacle-rich environments in this article. A novel graph-based motion planner is proposed. For a given workspace with a MMRS, valid configurations of the system are identified, where both formation and obstacle constraints are satisfied. Then, the whole valid configuration space is represented by an undirected graph, where vertices represent valid configurations and edges represent the connectivity between configurations. Afterward, efficient graph search algorithms can be directly used for planning, specifically, starting with the breadth-first search (BFS) algorithm to answer the question of whether there is a feasible path for the system. If there is, edges on the graph will update based on a cost function considering path length and formation preference, and the Dijkstra algorithm is further adopted to get the optimal path. In addition, a boundary densification method is proposed to refine the graph for better searching accuracy. Simulation results in various obstacle-rich environments demonstrate the effectiveness and generality of the proposed planner, which can be used to handle different formation constraints brought by the tasks.

The main contributions of this article are threefold. First, the valid configuration space of a MMRS is identified and mapped to an undirected graph, where both formation and obstacle

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constraints can be satisfied simultaneously. Second, efficient graph search algorithms can be directly used to determine whether there is a feasible path. Furthermore, the optimal path can be planned based on a designed cost function. Third, to improve the searching accuracy and speed after discretization, a boundary densification method is proposed to refine the graph.

The rest of this article is organized as follows. In Section II, some related work is discussed. In Section III, we define the basic concepts of multirobot motion planning problems. In Section IV, the proposed motion planner is explained in detail. In Section V, the planner is employed for different formation constraints by a case study in object transportation. The simulations and results are given in Section VI. Finally, Section VII concludes this article.

II. Related Work

In many studies of multirobot motion planning, formation constraints among robots are not the key point. For each robot, it regards other robots as dynamic obstacles, and thus collaboration is not considered [6], [7], [8]. However, as tasks become more complex, the collaboration between multiple robots is inevitable, especially for formation constraints.

Generally, approaches in the previous works on motion planning of MMRS under formation constraints can be divided into three categories. The first category is individually planning each robot while satisfying formation constraints and avoiding obstacles. Under the combination of the two behaviors, the system can hold desired formations [9]. Based on the leader–follower and potential-based methods, desired formations with obstacle avoidance were achieved in [10], [11], [12]. By combining the virtual structure and obstacle avoidance methods, Rezaee and Abdollahi [13] introduced an approach for multiple robots to maintain formations. Similarly, the combination can be found in [14]. Besides, learning-based methods are also employed for the problems. Formations could be adjusted independently by each robot to pass through obstacle areas based on reinforcement learning [15], [16]. In [17], the task space, including formation constraints and obstacle avoidance, was decomposed into different convex regions through a global planner. Then, configurations were planned through a local planner. In the above research, motion planners usually integrate two parts: avoiding obstacles and forming formations. This category of approaches can maintain desired formations tightly in the absence of obstacles. However, when facing obstacles, the deformation of formations is unpredictable due to the imposition of obstacle avoidance behavior. There is always a conflict between these two behaviors. Moreover, adjusting weights between them is also a problem with these approaches.

The second category considers the multirobot system as a whole 2-D area, which is planned to be strictly separated from obstacles. A system outlined rectangle approach was proposed in [18], regarding the multirobot system as a virtual rectangle to avoid obstacles. Then, the problem was simplified to the motion planning of a single robot. The idea can also be found in [19] and [20]. In [21] and [22], a constrained optimization method was presented for motion planning of multiple robots. The robots could always form the desired formations in the largest convex obstacle-free region computed in the neighborhood of the system. A region-based framework for multirobot systems was introduced in [23]. Based on virtual structure, robots moved and formed formations inside a changeable circular region always separated from obstacles. In [24], formation control and change were achieved by caging behaviors. The whole system was planned by the rapidly exploring random tree for obstacle avoidance. This category of approaches is well suited for some systems in performing special tasks, as in [25], [26], [27], [28]. However, in most cases, there are no explicit requirements to consider the system as a whole. Consequently, these approaches are overconstrained, resulting in partial loss of obstacle avoidance capability. Since robots can actually be separated, the system could have crossed obstacles instead of merely bypassing them.

It is noted that the first two categories do not deal with the coupling of different constraints very appropriately. In the third category, configurations of multirobot systems are first planned. For example, in order to handle objects through robots, desired configurations of the system had been defined and planned first in [29]. Their experiments showed the constraints could be satisfied. Based on generalized connectivity maintenance (GCM) theory [30] and transition-based rapidly exploring random tree (T-RRT) method [31], multirobot motion planning in clutter environments was achieved in [32]. Besides obstacle constraints, they additionally considered connectivity constraints of the system, which could be regarded as soft formation constraints. In their method, configurations that meet the connectivity requirements defined by GCM were planned first. Simulation results indicated that their method could be applied in obstacle-rich environments with different connectivity requirements. These works successfully demonstrate that first planning configurations can cope well with various constraints. However, these works do not directly consider the combination of formation and obstacle constraints, which is the focus of this article.

In multirobot motion planning studies, graph theory is a powerful tool, which provides natural abstractions about the information in a multiagent system [33], [34], [35], [36]. In [7], based on a graph and spanning tree representation, a multiphase approach was described. Through the relationship between multirobot planning problems and multiflow problems, integer linear programming models were proposed to compute optimal paths for multiple robots on connected graphs [37]. Their models were further extended in [38] in connection with different minimization objectives in path planning. The optimal solution or approximate optimal solution could be quickly calculated on the connected graphs. In [39], goal allocation and motion planning were achieved on roadmaps for nonholonomic robots. Among the three categories of work previously introduced, graph theory can also be found in [10], [12], [14], [23], [30], [32].

In the above studies containing graph theory, there are two main features. First, constraints within the multirobot system are not considered, such as formation or connectivity constraints. The only connection among multiple robots is the conflict between planned waypoints [40], [41], [42], [43]. Second, when considering constraints, the graph is usually used to describe

TABLE I
DEFINITION OF SOME IMPORTANT NOTATIONS

$\boldsymbol{r}_i$	The position of ith mobile robot
$\boldsymbol{p}$	The center of a multimobile robot system.
$\mathcal{P}$	The plane where mobile robots are located.
$\mathcal{S}$	Formations (shapes) of the MMRS satisfying formation constraints.
θ	The angle of a given formation (shape).
$\mathcal{C}$	Configurations of the MMRS with formation constraints.
$\boldsymbol{c}$	A configuration $\boldsymbol{c} \in \mathcal{C}$.
$\boldsymbol{c}^s$	A configuration for a given formation s, $\boldsymbol{c}^s = \{x, y, \theta\}$.
$\boldsymbol{o}_i$	The position of ith obstacle.
$\mathcal{W}(\boldsymbol{o}_i)$	Space occupied by obstacle $\boldsymbol{o}_i$.
$\mathcal{W}(\boldsymbol{r}_i)$	Space occupied by robot $\boldsymbol{r}_i$.
$\mathcal{C}_{\text{free}}$	Obstacle-free configurations, $\mathcal{C}_{\text{free}} \subset \mathcal{C}$.
$\mathcal{C}^s_{\text{free}}$	Obstacle-free configurations for a given formation s.
$\partial\mathcal{C}^s_{\text{free}}$	The boundary of $\mathcal{C}^s_{\text{free}}$.
$\mathcal{N}(\boldsymbol{c}^s, \xi)$	Neighborhood area of $\boldsymbol{c}^s$ with radius ξ.
$\boldsymbol{\Omega}(\boldsymbol{c}^s, \boldsymbol{p})$	Rotation angles for a given $\boldsymbol{c}^s$ and $\boldsymbol{p}$.

the internal connectivity characteristics such as the communication network between robots [36]. Specifically, vertices on the graph represent positions of robots while edges represent the distance or connectivity between robots. Then, costs can be defined using the graph. However, when planning motions of robots in a specific environment, it is not easy to achieve global planning and usually requires a combination with other methods, such as leader–follower method [10], [12] or virtual structure method [14], [23].

III. PRELIMINARIES

A. Problem Formulation

In this work, we consider multimobile robot motion planning under both obstacle-rich environments and formation constraints. The mobile robots are assumed to be holonomic, and they move on a 2-D plane.

Some important notations are listed in Table I for convenience. In a workplace with a MMRS $\mathcal{W} \subset \mathbb{R}^3$, the position of the ith mobile robot is denoted as $\boldsymbol{r}_i \in \mathbb{R}^2, i = 1, \ldots, n$, and n is used to denote the number of mobile robots. The plane where robots are located is denoted by $\mathcal{P} \subset \mathbb{R}^2$. The center of multirobot systems is denoted as $\boldsymbol{p} = [x, y] := \sum_{i=1}^{n} \frac{\boldsymbol{r}_i}{n} \in \mathbb{R}^2$, which is used to describe the location of the MMRS in the workplace. A formation (shape) s formed by multiple mobile robots is defined as follows:

$$s := \{d^*_{ij}\} \quad i, j = 1, \ldots, n \quad i \neq j$$
$$d^*_{ij} = \|\boldsymbol{r}_i - \boldsymbol{r}_j\|_2 \tag{1}$$

which consists of the relative distance between robots. Hereafter, we use formation as the overall shape formed by a multirobot system. This definition of formations is very common in related studies [12], [13], [14]. We further use $\theta \in [0, 2\pi)$ to indicate the orientation of a formation. As shown in Fig. 1, it can be seen that the two formations are the same, but the robots are in different locations. Therefore, an angle is needed to distinguish them. In order to prevent arbitrary selection of orientation in a formation, θ is defined as the angle of $\boldsymbol{p}\boldsymbol{r}_1$ with the x-axis. To sum up, a configuration $\boldsymbol{c}$ of the MMRS can be described as $\boldsymbol{c} = \{\boldsymbol{p}, \theta, s\}$. Once $\boldsymbol{c}$ is given, the position of each robot $\{\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n\}$ can be determined accordingly.

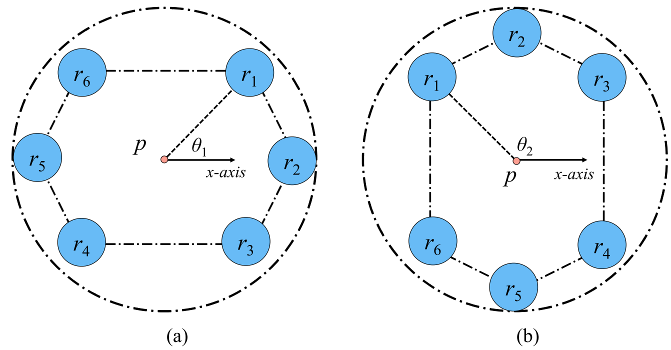

Fig. 1. Same formation with different angles. $\boldsymbol{p}$ is the same in (a) and (b), while positions of robots are different. (a) Shape with θ_1. (b) Shape with θ_2.

Now, let $\boldsymbol{o}_i \subset \mathbb{R}^2, i = 1, \ldots, m$ denote the position of the ith obstacle, and m is the number of obstacles. $\mathcal{W}(\boldsymbol{o}_i) \subset \mathbb{R}^3$ is used to denote the workspace occupied by the ith obstacle. Similarly, $\mathcal{W}(\boldsymbol{r}_i)$ is the workspace occupied by the ith robot.

Definition 1: The obstacle-free configuration space $\mathcal{C}_{\text{free}}$ of the MMRS is satisfying

$$\mathcal{C}_{\text{free}} := \{\boldsymbol{c} \mid \mathcal{R} \cap \mathcal{O} = \varnothing\}$$
$$\text{where } \mathcal{R} = \mathcal{W}(\boldsymbol{r}_i) \cup \cdots \cup \mathcal{W}(\boldsymbol{r}_n)$$
$$\mathcal{O} = \mathcal{W}(\boldsymbol{o}_i) \cup \cdots \cup \mathcal{W}(\boldsymbol{o}_m). \tag{2}$$

In other words, $\boldsymbol{c} \in \mathcal{C}_{\text{free}}$ is called a valid configuration, which indicates that the MMRS satisfies the formation constraints, and all robots are not in contact with obstacles. Thus, if we plan in $\mathcal{C}_{\text{free}}$, both formation and obstacle constraints can be satisfied simultaneously. If we directly plan the motion of each robot to maintain formation and avoid obstacles, it is hard to satisfy both constraints. Moreover, it will suffer from the curse of dimensionality, which is as high as $\mathbb{R}^{2n}$.

In specific applications of a MMRS, the workspace $\mathcal{W}$ with the information of obstacles $\mathcal{O}$ is given. Formations $\mathcal{S}$ are determined by the user or the system itself, and examples will be given in Section V. The system is required to move from the initial configuration $\boldsymbol{c}_0$ to the goal configuration $\boldsymbol{c}_g$. Thus, the planning problem considered in this article can be formulated as follows:.

$$\text{Input}: \ \mathcal{W}, \mathcal{O}, \mathcal{S}, \boldsymbol{c}_0, \boldsymbol{c}_g$$
$$\text{Output}: \ \text{An optimal trajectory in } \mathcal{C}_{\text{free}} \tag{3}$$

To solve (3), we need to build the valid configuration space $\mathcal{C}_{\text{free}}$, and then find the optimal trajectory in $\mathcal{C}_{\text{free}}$.

B. Problem Analysis

Definition 2: Two configurations are connected if they can be switched to each other through all available configurations in

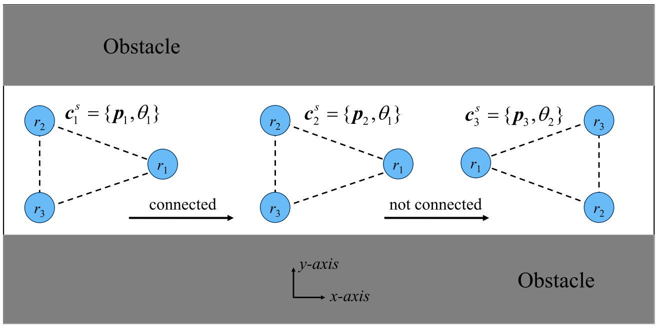

Fig. 2. Different valid configurations in $\mathcal{C}_{\text{free}}^s$ of a system composed of three mobile robots.

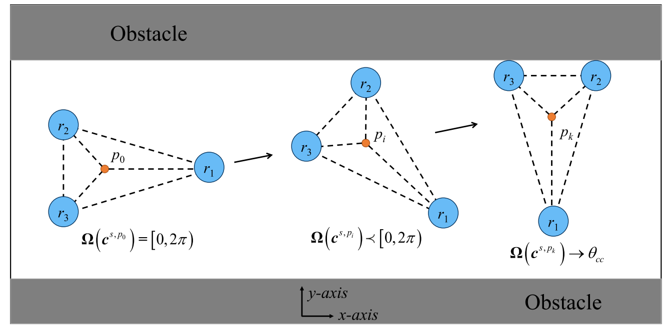

Fig. 3. Change of $\mathbf{\Omega}$ when approaching the boundary $\partial\mathcal{C}_{\text{free}}^s$ of valid configuration space $\mathcal{C}_{\text{free}}^s$.

$\mathcal{C}_{\text{free}}$. In other words, there exists a trajectory in $\mathcal{C}_{\text{free}}$ connecting them.

In order to design the planner, we first investigate the properties of the valid configuration space. We start with the case that does not contain formation switch. A configuration for the given formation s is denoted as $\boldsymbol{c}^s = \{x, y, \theta\}$, and valid configuration space for s is defined as $\mathcal{C}_{\text{free}}^s$. As shown in Fig. 2, there are three valid configurations for a given formation s. $\boldsymbol{c}_1^s$ and $\boldsymbol{c}_2^s$ are connected according to Definition 2, since the system can move from $\boldsymbol{c}_1^s$ to $\boldsymbol{c}_2^s$ while keeping the formation unchanged. However, $\boldsymbol{c}_2^s$ and $\boldsymbol{c}_3^s$ are not connected since $\boldsymbol{c}_2^s$ cannot switch to $\boldsymbol{c}_3^s$ while maintaining the formation, although they are both in $\mathcal{C}_{\text{free}}^s$. Hence, it is necessary to study the connectivity between different valid configurations for planning.

Definition 3: The neighborhood area $\mathcal{N}(\boldsymbol{c}^s, \xi) := \{\tilde{\boldsymbol{c}}^s\}$ of a configuration $\boldsymbol{c}$ is

$$\begin{aligned} \left\|(\tilde{x} - x)^2 + (\tilde{y} - y)^2\right\|_2 &< \xi \\ \text{abs}(\tilde{\theta} - \theta) &< \xi, \ \xi > 0 \\ \text{where } \tilde{\boldsymbol{c}}^s &= \{\tilde{x}, \tilde{y}, \tilde{\theta}\}. \end{aligned} \tag{4}$$

Theorem 1: For any $\boldsymbol{c}^s \in \mathcal{C}_{\text{free}}^s$, $\boldsymbol{c}^s$ is connected in a neighborhood area $\mathcal{N}(\boldsymbol{c}^s, \xi)$.

Proof: $\forall \boldsymbol{c}^s = \{x, y, \theta\} \in \mathcal{C}_{free}^s$, from Definition 1, all robots are separated from obstacles although their distances may be very small. In this case, a specific obstacle-free space $\mathcal{T}_i$ can be found for the ith robot, which satisfies

$$\mathcal{W}(\boldsymbol{r}_i) \subset \mathcal{T}_i, \mathcal{T}_i \cap \mathcal{O} = \varnothing. \tag{5}$$

$\mathcal{T}_i$ is a bounded open set. Since (5) is available for each robot, the whole formation can move or rotate freely in any direction if $\mathcal{W}(\tilde{\boldsymbol{r}}_i)$ still stays in $\mathcal{T}_i$ after movement, which will generate new valid configurations. Therefore, $\boldsymbol{c}^s$ is connected with these configurations, and the area is denoted as $\mathcal{N}(\boldsymbol{c}^s, \xi)$. Obviously, ξ is strongly related to $\mathcal{T}_i$ and is relatively bigger in an environment without obstacles. It should be pointed out that Definition 3 is a little conservative because the whole formation may be able to move bigger than ξ in some directions. However, the conservative definition does not affect overall connectivity since the movement of whole formation can be repeated many times. ■

Two valid configurations in $\mathcal{C}_{\text{free}}^s$ may not be connected because the space $\mathcal{C}_{\text{free}}^s$ always has boundary due to obstacles. If we specify the boundary $\partial\mathcal{C}_{\text{free}}^s$ of $\mathcal{C}_{\text{free}}^s$, it will be more convenient to verify the connectivity between configurations and plan the trajectory. Now, we use $\mathbf{\Omega}(\boldsymbol{c}^{s,\boldsymbol{p}})$ to denote rotation angles for a given formation s and $\boldsymbol{p}$, and the maximum range of $\mathbf{\Omega}(\boldsymbol{c}^{s,\boldsymbol{p}})$ is $[0, 2\pi)$.

Definition 4: Critical contact configuration $\boldsymbol{c}_{cc}$ denotes the situation that

$$\begin{gathered} \boldsymbol{c}_{cc} \not\subset \mathcal{C}_{\text{free}}^s, \forall \xi > 0, \text{ satisfying} \\ \mathcal{N}(\boldsymbol{c}_{cc}, \xi) \not\subset \mathcal{C}_{\text{free}}^s \text{ and } \mathcal{N}(\boldsymbol{c}_{cc}, \xi) \not\subset \mathcal{C}^s \backslash \mathcal{C}_{\text{free}}^s \end{gathered} \tag{6}$$

It indicates that in this configuration, the robots come into contact with obstacles without overlapping. And $\boldsymbol{c}_{cc} = \{x, y, \theta\}$, θ is defined as the critical contact angle θ_{cc}.

Theorem 2: If critical contact cannot be maintained during the rotation of the formation, then $\boldsymbol{c}^s$ approaches the boundary $\partial\mathcal{C}_{free}^s$ if and only if $\mathbf{\Omega}(\boldsymbol{c}^{s,\boldsymbol{p}})$ reduces or converges to a θ_{cc}. Moreover, the angle of the formation is θ_{cc} when $\boldsymbol{c}^s \in \partial\mathcal{C}_{free}^s$.

Proof: Denote the initial position $\boldsymbol{p} = \boldsymbol{p}_0$ and $\boldsymbol{c}^{s,\boldsymbol{p}_0} \in \mathcal{C}_{free}^s$. According to Theorem 1, $\boldsymbol{c}^{s,\boldsymbol{p}_0}$ is connected in $\mathcal{N}(\boldsymbol{c}^{s,\boldsymbol{p}_0}, \xi_0)$, $\boldsymbol{c}^{s,\boldsymbol{p}_1}$ is randomly chosen in $\mathcal{N}(\boldsymbol{c}^{s,\boldsymbol{p}_0}, \xi_0)$, then do

$$(\boldsymbol{p}_{k+1} - \boldsymbol{p}_k) = \lambda_k(\boldsymbol{p}_1 - \boldsymbol{p}_0) \quad \boldsymbol{c}^{s,\boldsymbol{p}_{k+1}} \in \mathcal{N}(\boldsymbol{c}^{s,\boldsymbol{p}_k}, \xi_k) \tag{7}$$

where λ_k is a positive coefficient, and this moves the system in one direction. During (7), $\mathcal{N}(\boldsymbol{c}^{s,\boldsymbol{p}_k}, \xi_k)$ may become bigger or smaller depending on the environment. $\boldsymbol{c}^{s,\boldsymbol{p}_{k+1}}$ approaches the boundary $\partial\mathcal{C}_{\text{free}}^s$ if and only if $\xi_{k+1} \to 0$. That means, $\mathcal{T}_i \to \mathcal{W}(\boldsymbol{r}_i)$ at least in one direction for one robot and the robot is fairly close to obstacles. The configuration will then transition into a critical contact configuration $\boldsymbol{c}_{cc} = \{\boldsymbol{p}, \theta_{cc}, s\}$, and $\boldsymbol{c}_{cc} \in \partial\mathcal{C}_{\text{free}}^s$. Due to the assumption, the formation cannot rotate around $\boldsymbol{p}$. Consequently, $\mathbf{\Omega}(\boldsymbol{c}^{s,\boldsymbol{p}})$ reduces since some angles cause collision, and then finally converges to one single angle. In this process, boundaries of $\mathbf{\Omega}(\boldsymbol{c}^{s,\boldsymbol{p}})$ are all critical contact angles. A case is shown in Fig. 3, the system forms a rigid formation, where $\mathbf{\Omega}(\boldsymbol{c}^{s,\boldsymbol{p}_0}) = [0, 2\pi)$ and $\mathbf{\Omega}(\boldsymbol{c}^{s,\boldsymbol{p}_k}) \to \theta_{cc}$. It can be seen that $\mathbf{\Omega}$ gradually reduces to one critical contact angle when approaching the boundary. ■

Theorems 1 and 2 are enrolled for only one formation. But in some situations, the formation is expected or required to switch. As shown in Fig. 4, the system can never pass the obstacle if the

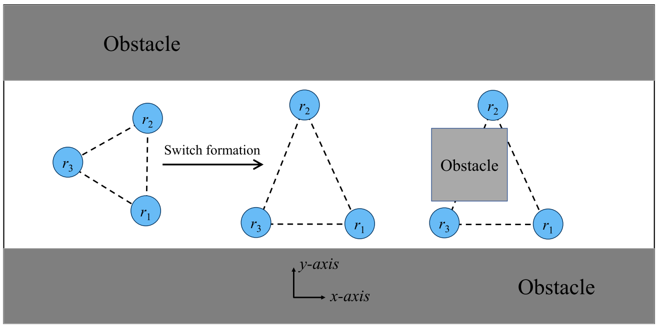

Fig. 4. System is forced to switch formation in order to pass obstacles.

formation remains unchanged. Therefore, whether two configurations in different formations, $\boldsymbol{c}^{s_1,\boldsymbol{p}} \in \mathcal{C}_{\text{free}}^{s_1}$ and $\boldsymbol{c}^{s_2,\boldsymbol{p}} \in \mathcal{C}_{\text{free}}^{s_2}$ are connected also needs to be studied in order to enhance the connectivity of configuration space.

Definition 5: $\boldsymbol{c}^{s_1,\boldsymbol{p}} \in \mathcal{C}_{\text{free}}^{s_1}$ and $\boldsymbol{c}^{s_2,\boldsymbol{p}} \in \mathcal{C}_{\text{free}}^{s_2}$ are connected in the same $\boldsymbol{p}$ if satisfying: 1) $\forall i$, the ith robot can move from $\boldsymbol{r}_i^{s_1}$ to $\boldsymbol{r}_i^{s_2}$ without collision with obstacles; 2) $\forall i, j$, in any moment t during formation switch, $\|\boldsymbol{r}_i(t) - \boldsymbol{r}_j(t)\|_2 \leq \|\boldsymbol{r}_i^{s_1} - \boldsymbol{r}_j^{s_1}\|_2$ or $\|\boldsymbol{r}_i(t) - \boldsymbol{r}_j(t)\|_2 \leq \|\boldsymbol{r}_i^{s_2} - \boldsymbol{r}_j^{s_2}\|_2$.

The first condition is to ensure safety while the second is to make robots satisfy formation constraints.

Theorem 3: For $\boldsymbol{c}^{s_1,\boldsymbol{p}} \in \mathcal{C}_{free}^{s_1}$, $\boldsymbol{c}^{s_2,\boldsymbol{p}} \in \mathcal{C}_{free}^{s_2}$, if robots start and stop moving at the same time and collisions between robots are prevented, and $\forall i$, $\exists$ convex obstacle-free region $\mathcal{T}_i$, satisfying $\mathcal{W}(\boldsymbol{r}_i^{s_1}) \subset \mathcal{T}_i$ and $\mathcal{W}(\boldsymbol{r}_i^{s_2}) \subset \mathcal{T}_i$, $\boldsymbol{c}^{s_1,\boldsymbol{p}}$ and $\boldsymbol{c}^{s_2,\boldsymbol{p}}$ are connected.

Proof: For the first condition, according to the properties of convex regions, the connecting line of any two points in $\mathcal{T}_i$ must be in $\mathcal{T}_i$. Therefore, $\mathcal{W}(\boldsymbol{r}_i(t)) \subset \mathcal{T}_i$ if $\boldsymbol{r}_i(t)$ is in the straight line path from $\boldsymbol{r}_i^{s_1}$ to $\boldsymbol{r}_i^{s_2}$. Furthermore, when taking this path, the second condition is also satisfied. It can be derived from a more general situation as follows:

$$\begin{aligned}
&\exists \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{a}^*, \boldsymbol{b}^* \in \mathbb{R}^N, \ \forall \eta \in [0,1] \\
d =\ & \|[\eta(\boldsymbol{a} - \boldsymbol{a}^*) + \boldsymbol{a}^*] - [\eta(\boldsymbol{b} - \boldsymbol{b}^*) + \boldsymbol{b}^*]\| \\
=\ & \|\eta(\boldsymbol{a} - \boldsymbol{b}) + (1-\eta)(\boldsymbol{a}^* - \boldsymbol{b}^*)\| \\
\leq\ & \eta \|\boldsymbol{a} - \boldsymbol{b}\| + (1-\eta)\|\boldsymbol{a}^* - \boldsymbol{b}^*\| \\
\leq\ & \max\{\|\boldsymbol{a} - \boldsymbol{b}\|, \|\boldsymbol{a}^* - \boldsymbol{b}^*\|\}. \qquad (8)
\end{aligned}$$

In our case, for any two robots i, j, let $\boldsymbol{a} \leftarrow \boldsymbol{r}_i^{s_1}$, $\boldsymbol{b} \leftarrow \boldsymbol{r}_j^{s_1}$, $\boldsymbol{a}^* \leftarrow \boldsymbol{r}_i^{s_2}$, $\boldsymbol{b}^* \leftarrow \boldsymbol{r}_j^{s_2}$. Since we assume that robots start and stop moving at the same time, $[\eta(\boldsymbol{a} - \boldsymbol{a}^*) + \boldsymbol{a}^*] \leftarrow \boldsymbol{r}_i(t)$ and $[\eta(\boldsymbol{b} - \boldsymbol{b}^*) + \boldsymbol{b}^*] \leftarrow \boldsymbol{r}_j(t)$. Therefore, according to (8), the distance $d_{ij}(t)$ between the two robots

$$\begin{aligned}
d_{ij}(t) &= \|\boldsymbol{r}_i(t) - \boldsymbol{r}_j(t)\|_2 \\
&\leq \max\left\{\|\boldsymbol{r}_i^{s_1} - \boldsymbol{r}_j^{s_1}\|_2, \|\boldsymbol{r}_i^{s_2} - \boldsymbol{r}_j^{s_2}\|_2\right\}. \qquad (9)
\end{aligned}$$

During movement, $d_{ij}(t)$ will not exceed the distance of two robots in s_1 or s_2. The second condition is also satisfied. Then, $\boldsymbol{c}^{s_1,\boldsymbol{p}} \in \mathcal{C}_{\text{free}}^{s_1}$ and $\boldsymbol{c}^{s_2,\boldsymbol{p}} \in \mathcal{C}_{\text{free}}^{s_2}$ are connected. ■

Remark 1: Theorem 3 relies on two conditions. The first is that robots start and stop moving during formation switch, which is a fundamental consideration in the collaboration. The second is that collisions between robots are prevented. In many multirobot cooperative tasks, the distribution of robots within the formation is required to be dispersed and uniform for a better sharing of the workload. Therefore, there is usually plenty of room between different robots. Thus, robot–robot collisions can be avoided during formation switch.

Remark 2: Theorem 3 provides only a proof of sufficiency for Definition 5. In the problem considered in this article, the two conditions can be satisfied. For the more general case of Theorem 3, such as how to generate collision-free trajectories for robots, some references can be found in [44].

In the above statement, we adequately introduce the properties of valid configuration space. The connectivity between configurations for a given formation has been studied in Theorems 1 and 2, while the switch between different formations has been studied in Theorem 3. These can be used to build the $\mathcal{C}_{\text{free}}$ and plan the trajectory, based on which, our planner is designed.

IV. Proposed Planner

The focus of this article is on multimobile robot planning in the obstacle-rich environment, which is assumed to be static and globally known in advance. The framework of our planner is shown in Fig. 5. Since it is extremely hard to obtain the continuous case of $\mathcal{C}_{\text{free}}$ under complex formation and obstacle constraints, we will solve the problem by using the discretization method. For a given workspace with a MMRS, the configuration space is firstly discretized into cells. Then, the $\mathcal{C}_{\text{free}}$ is mapped into a graph, as the blue points shown in Fig. 5. Each vertex represents a valid configuration, which contains position (x, y) and orientation θ of a formation s in the workspace. Then, the edges are generated by checking the connectivity between configurations in $\mathcal{C}_{\text{free}}$, based on which an undirected graph is built. The BFS method [45] can be applied to quickly decide whether a feasible path exists or not. If the answer is yes, a cost function based on the path lengths of robots and formation preference is assigned to update the graph. Finally, the Dijkstra method [46] can be applied to find the optimal path on the updated graph. Our planner is able to deal with different formation constraints in obstacle-rich environments, which will be discussed in the next section.

A. Mapping of the Configuration Space

The first step to solve (3) is to find the $\mathcal{C}_{\text{free}}$. In this article, $\mathcal{C}_{\text{free}}$ is built and mapped to an undirected graph. For a given workspace $\mathcal{W}$ with the plane $\mathcal{P}$, configurations are identified through a relationship $\mathcal{M}(\cdot)$ as follows:

$$\begin{aligned}
\mathcal{M}(\boldsymbol{p}) &:= \{\boldsymbol{c} \mid x_{\boldsymbol{c}} = x_{\boldsymbol{p}}, y_{\boldsymbol{c}} = y_{\boldsymbol{p}}, \boldsymbol{c} \in \mathcal{C}\} \\
\mathcal{M}(\boldsymbol{c}) &:= \{\boldsymbol{p} \mid x_{\boldsymbol{p}} = x_{\boldsymbol{c}}, y_{\boldsymbol{p}} = y_{\boldsymbol{c}}, \boldsymbol{p} \in \mathcal{P}\}. \qquad (10)
\end{aligned}$$

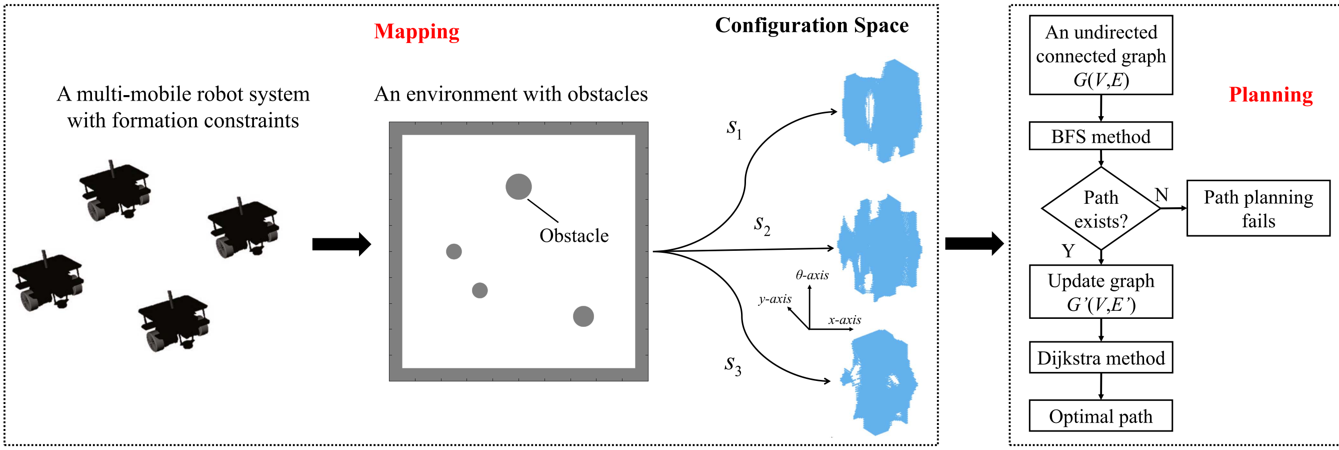

Fig. 5. General framework of our planner. The valid configuration space $\mathcal{C}_{\text{free}}$ is built by using the graph with boundary densification. Each vertex in $\mathcal{C}_{\text{free}}$ contains position (x, y) and orientation θ of a formation in the workspace. Edges of the graph are generated based on the connectivity of valid configurations. Whether a path exists is quickly determined on the graph. The optimal path is planned on the updated graph considering path length and formation preference. Then, motions of robots are computed on the path.

In the MMRS, a configuration is defined as $\boldsymbol{c} = \{x, y, \theta, s\}$. Since the dimension of $\boldsymbol{c}$ is higher than $\boldsymbol{p}$, other two parameters θ and s are needed in addition to $\mathcal{M}(\boldsymbol{p})$, which denote the orientation and formation of the system, respectively. It is noted that under complex formation and obstacle constraints, it is almost impossible to explicitly describe the continuous configuration space. Thus, we solve the problem by the discretization method. The plane $\mathcal{P}$ and rotation angles $\boldsymbol{\Omega}$ are first discretized. Consequently, for a given formation s, the 3-D valid configuration space $\mathcal{C}^s_{\text{free}}$ is composed of finite discretized points due to the limited environment and angle range. As for formations $\mathcal{S}$, they are determined by the user or the system itself, which also contain finite elements in general. Sometimes, two configurations in different formations may not be connected, which will affect the subsequent planning. In this case, one solution is to relax the formation constraints by setting more formations in $\mathcal{S}$ to reduce the differences between formations. Therefore, the whole valid configuration space $\mathcal{C}_{\text{free}}$ is finite after discretization, and we use a graph to represent the $\mathcal{C}_{\text{free}}$.

Now, $\mathcal{P}$ and $\boldsymbol{\Omega}$ are needed to be discretized. Specifically, $\mathcal{P}$ is discretized by g while $\boldsymbol{\Omega} \subset [0, 2\pi)$ is discretized by α. g, and α are discrete scales, which determine the success rate and mapping time of the planner. On the one hand, small enough discrete scales will bring about a good enough approximation of the continuous configuration space. On the other hand, smaller discrete scales will result in a larger computational cost. To solve the problem practicably, we need to choose a suitable range for them. Generally, they are strongly affected by the control accuracy of robots and size of obstacles. g is chosen according to the following rules.

(1) $g \in [g_{\min}, g_{\max}]$.

(2) $g_{\min}$ is related to computational cost and accuracy. The control accuracy of robots is limited and can never reach such a resolution as small as possible. Therefore, $g_{\min}$ cannot be selected too small. Otherwise, it will be impractical or computationally unacceptable.

(3) $g_{\max}$ is related to the connectivity of configurations. This is decided by the size of obstacles and robots. Path planning may fail if $g_{\max}$ is selected too big.

The choice of α is similar to g. It cannot be too small because of the accuracy, nor can it be too big to cross obstacles during rotation. Once discrete scales are determined, for each formation s, the 3-D configuration space $\mathcal{C}^s_{\text{free}}$ can be built. In order to refine the configuration space after discretization, a boundary densification method is proposed. Valid configurations near the boundary $\partial\mathcal{C}^s_{\text{free}}$ are selected. Then, $\boldsymbol{p}$ near the boundary in $\mathcal{P}$ is identified by $\mathcal{M}(\boldsymbol{c})$. For areas on $\mathcal{P}$ near the boundary, the minimal discrete scale $g_{\min}$ will be used to discretize them further. The newly resulting $\boldsymbol{p}$ is added to construct the configuration space. In this way, the most realistic situation of $\mathcal{C}^s_{\text{free}}$ can be approximated to the greatest extent without excessively increasing the computation.

Fig. 6 provides an example of our proposed boundary densification method. The system maintains a given formation s in the workspace, and the valid configuration space $\mathcal{C}^s_{\text{free}}$ is a 3-D space as previously illustrated. Fig. 6(a) shows the workspace and valid configuration space. For better visualization, we exhibit only some parts of the 3-D valid configuration space. Fig. 6(b) illustrates the effect of boundary densification, specifically, in the part of valid configuration space where $\theta = 0$. In the valid configuration space, the blue points are generated by $\mathcal{M}(\boldsymbol{p})$ of points $\boldsymbol{p}$ discretized by g. After finding the areas near $\partial\mathcal{C}^s_{\text{free}}$, the workspace undergoes further discretization by $g_{\min}$, generating some new points, which are transformed into the red points by $\mathcal{M}(\cdot)$ in $\mathcal{C}^s_{\text{free}}$. It can be seen that double discretization can increase the density of boundary, thereby increasing the success rate of planning, which will be demonstrated by simulations later.

The mapping process is detailed in Algorithm 1, where some key functions are explained as follows:

1) `PlaneDiscretized` and `AngleDiscretized` discretize the workspace by the given discrete scales.

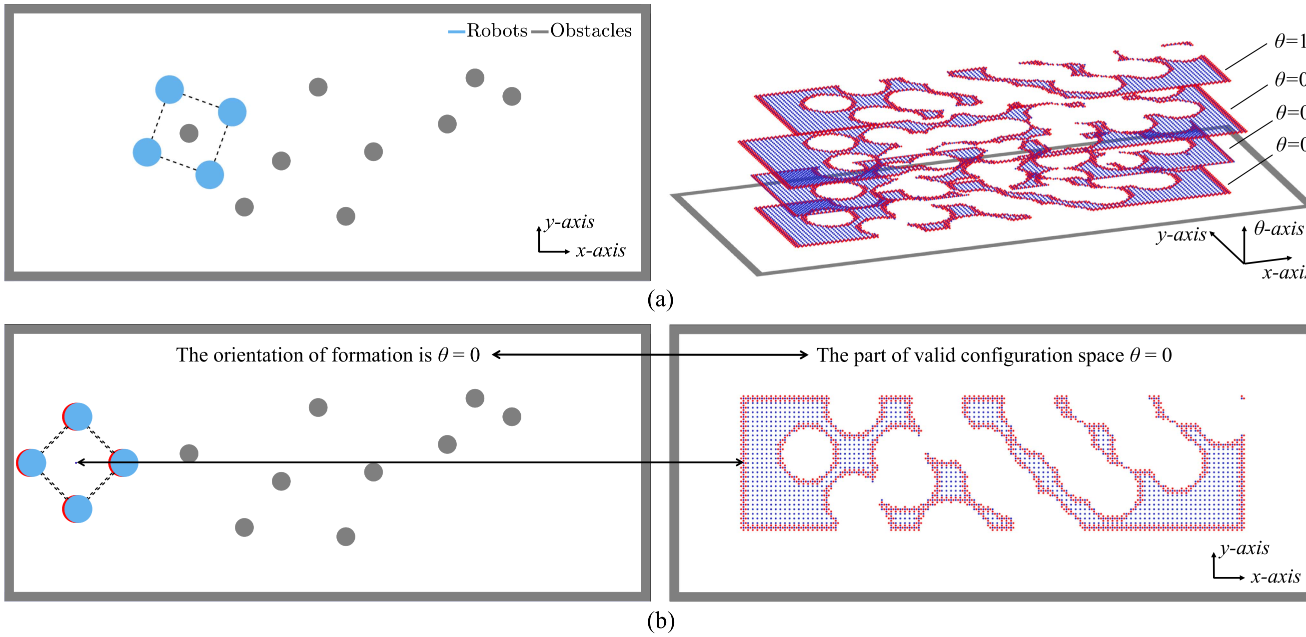

Fig. 6. Example with boundary densification. The workspace is firstly discretized by the scale g, generating the valid configuration space composed of blue points. Subsequently, areas in proximity to the boundary are identified and further discretized by $g_{\min}$, generating the valid configuration space composed of red points. (a) The workspace and valid configuration space. For better visualization, we only show parts of the 3-D valid configuration space. (b) The effect of boundary densification method. It refines the situations when configurations approach the boundary.

Algorithm 1: Configurations Mapping.

Input: Workspace $\mathcal{W}$; Obstacles $\mathcal{O}$; Formations $\mathcal{S}$; Discrete scales g, α, $g_{\min}$.

Output: Valid configuration space $\mathcal{C}_{free}$.

1 Initializing: $\mathcal{C}_{free} \leftarrow \varnothing$;
2 $\boldsymbol{p}_i \leftarrow$ `PlaneDiscretized`(g);
3 $\theta \leftarrow$ `AngleDiscretized`(α);
4 **foreach** $\boldsymbol{p}_i$ **do**
5   `DenoteOrientation`(θ), $\theta \in \boldsymbol{\Omega}$
6   `DenoteFormation`(s), $s \in \mathcal{S}$
7     $\boldsymbol{c} \leftarrow \mathcal{M}(\boldsymbol{p}_i)$;
8   **if** `ValidConfiguration`$(\boldsymbol{c})$ is true **then**
9         $\boldsymbol{c}$ is added to $\mathcal{C}_{free}$;
10   **end**
11 **end**
12 **foreach** $\boldsymbol{c} \in \mathcal{C}_{free}$ **do**
13   **if** `BoundaryDetect`$(\boldsymbol{c})$ is true **then**
14         $\boldsymbol{p}_i \leftarrow \mathcal{M}(\boldsymbol{c})$;
15         $\boldsymbol{p}_{\text{new}} \leftarrow$ `BoundaryDiscretized`$(\boldsymbol{p}_i, g_{\min})$;
16   **foreach** $\boldsymbol{p}_{\text{new}}$ **do**
17   Repeat 5-10;
18   **end**
19   **end**
20 **end**

2) `DenoteFormation` and `DenoteOrientation` determine the formation and formation orientation for robots in the center $\boldsymbol{p}_i$. Then, the relationship between the workspace and configuration space can be built.
3) `ValidConfiguration` checks the positions of robots in a configuration $\boldsymbol{c}$. If robots and obstacles do not collide, as stipulated by Definition 1, then $\boldsymbol{c}$ is added to $\mathcal{C}_{\text{free}}$ for the planning process.
4) `BoundaryDetect` determines whether a configuration $\boldsymbol{c}$ is near the boundary based on Theorem 2, which is achieved by assessing the angle θ. If the adjacent configuration with angle θ is not valid, $\boldsymbol{c}$ is near the boundary, and the neighboring workspace will be further discretized by `BoundaryDiscretized`, as shown in Fig. 6(b).

B. Planning on the Graph

The second step to solve (3) is to plan the trajectory in $\mathcal{C}_{\text{free}}$. In this article, graphs are used to describe the valid configurations of system and the connectivity between valid configurations. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a graph, where $\mathcal{V} = \{v_i\}$ is the vertex set and $\mathcal{E} = \{e_{ij} := (v_i, v_j)\}$ is the edge set. Graphs can be divided into directed graphs and undirected graphs, depending on e_{ij} and e_{ji}. If $e_{ij} = e_{ji}$ for all edges, the graph is undirected. Otherwise, the graph is directed. For the problem we consider, when a configuration can switch to the other configurations, it is also reversible. So, the graph is always undirected. In path planning, valid configurations are stored in $\mathcal{V}$, and costs between them are stored in $\mathcal{E}$. Afterward, some efficient graph search algorithms can be directly applied.

We have obtained vertices $\mathcal{V}$ for the graph from $\mathcal{C}_{\text{free}}$ generated by Algorithm 1. Then, it needs to specify edges $\mathcal{E}$, which represent costs between valid configurations. To improve planning efficiency, whether two configurations are connected will be determined primarily, and the specific cost is not considered at first. The connectivity between configurations can be verified according to Theorems 1 and 3. Afterward, $\mathcal{E}$ can be obtained. Specifically, e_{ij} is added to the $\mathcal{E}$ if v_i and v_j are connected. In $\mathcal{V}$, the initial and goal configurations are denoted as v_{init} and v_{goal}, respectively. For $\mathcal{G}$, we can use the BFS method to quickly search whether there is a path from v_{init} to v_{goal}. If it exists, we will calculate the specific cost in $\mathcal{E}$ and then find the optimal path. If it does not exist, path planning fails, and all subsequent calculations will be saved. It should be noted that the graph might not be a connected graph, or may include multiple connected graphs and isolated vertices. Therefore, we first quickly determine if the two vertices are connected, and then search for the optimal path.

There are three options for configuration change: movement, rotation and formation switch. A cost function is designed based on the consideration of path lengths of robots and formation preference as follows:

$$e_{ij} = \begin{cases} \sum_{k=1}^{N} \lambda_s \left\| \boldsymbol{r}_k^{v_i} - \boldsymbol{r}_k^{v_j} \right\|_2 & \text{if move or rotate} \\ \sum_{k=1}^{N} \left\| \boldsymbol{r}_k^{v_i} - \boldsymbol{r}_k^{v_j} \right\|_2 & \text{if switch formation} \end{cases} \tag{11}$$

where $\sum_{k=1}^{N} \|\boldsymbol{r}_k^{v_i} - \boldsymbol{r}_k^{v_j}\|_2$ is the total distance traveled by all robots. $\lambda_s = \{\lambda_{s_1}, \lambda_{s_2}, \dots\}$. λ_s is the indicator of formation preference, which is different in different formations. A smaller λ_s will result in lower costs. Accordingly, the system will try to maintain the formation with smaller λ_s. We can assign priority between different formations by adjusting λ_s. It should be pointed out that the cost function is related to applications of the MMRS. In this article, it is designed to maintain the most preferred formation. If needed, the cost function can also be designed for maintaining the current formation or some other strategies. After calculating the specific cost, $\mathcal{G}$ is updated to $\mathcal{G}'$. For searching the optimal path in $\mathcal{G}'$, some existing methods can be used directly, such as the generic algorithm or Floyd–Warshall algorithm [47]. The framework of our planner is universal to different search algorithms. In this article, the Dijkstra algorithm is enrolled. Searching twice by different methods is trying to save time and computational costs.

The planning process is detailed in Algorithm 2, where some key functions are explained as follows.

1) `ConnectionDetect` verifies the connectivity between two configurations. If robots do not switch formations, it is implemented based on Theorem 1. Then, we can use the discrete scales to determine the neighborhood area and check the connectivity. If robots switch formations, it is implemented based on Theorem 3. In this process, it is important to identify the described obstacle-free space. Fortunately, there are several methods in existing work to find obstacle-free areas for robots, such as the virtual circles in [23] or the polytopes in [22]. In this article, we employ the method in [22], which can rapidly compute an obstacle-free polytope for robots in obstacle-rich environments.
2) `BFS` and `Dijkstra` apply the corresponding search methods on the graph to find a path between v_{init} and v_{goal}.

Algorithm 2: Planning on Graph.

Input: Valid configuration space $\mathcal{C}_{free}$; Cost function; Initial vertex v_{init} and goal vertex v_{goal}.
Output: Robot path $R_1, \dots, R_n$.

Initializing: $R_1, \dots, R_n \leftarrow \varnothing$, $\mathcal{G}(\mathcal{V}, \mathcal{E}) \leftarrow \varnothing$.
foreach $\boldsymbol{c}_i \in \mathcal{C}_{free}$ **do**
  $\boldsymbol{c}_i$ is added to $\mathcal{V}$ as v_i;
end
for $v_i, v_j \in \mathcal{V}$ **do**
  if `ConnectionDetect`(v_i, v_j) is true **then**
    e_{ij} is added to $\mathcal{E}$;
  end
end
$\Gamma \leftarrow$ `BFS`$(\mathcal{G}, v_{\text{init}}, v_{\text{goal}})$;
if Γ *exists* **then**
  $\mathcal{G}' \leftarrow$ `CostFunction`$(\mathcal{G})$;
  $\Gamma \leftarrow$ `Dijkstra`$(\mathcal{G}', v_{\text{init}}, v_{\text{goal}})$;
  $R_1, \dots, R_n \leftarrow$ `RobotPath`(Γ);
  return $R_1, \dots, R_n$;
else
  return $R_1, \dots, R_n$;
end

V. Case Study in Object Transportation

We take multirobot object transportation as a typical application of formation constraints. There are not only constraints brought by formations, but also by objects in transportation [48]. Therefore, it is more challenging than other scenarios. Generally, multirobot transportation systems are composed of mobile robots and the carrier. Carriers can be mobile robot itself [49], [50], [51], deformable sheet [52], [53], and manipulator [54], [55], [56], which result in different system model and formation constraints, and the first two will be used as examples in this article.

Now, a little modification is made to the previous definition. $\boldsymbol{p} = [x, y]$ is used to denote the position of object, and $\mathcal{W}(\boldsymbol{p})$ is used to denote the space occupied by object and carriers. There is a certain relationship as follows:

$$\boldsymbol{p} = \mathfrak{F}(\boldsymbol{r}_1, \dots, \boldsymbol{r}_n) \tag{12}$$

where $\mathfrak{F}$ is the mapping between positions of robots and the object, which is directly determined by the carrier. In fact, formation $\mathcal{S}$ are the internal relative information of multirobot systems. Therefore, if we only care about the relative positional relationship within the system, (12) can be rewritten as follows:

$$\hat{\boldsymbol{p}} = \mathfrak{F}(\mathcal{S}) \tag{13}$$

where $\hat{\boldsymbol{p}}$ is the relative position of objects inside the system. Generally, (12) is used to describe the absolute locations of the transportation system in the workplace, and (13) is used for relative positions within the system. Since there are new constraints brought by the object and carriers, the definition of obstacle-free configuration space (2) is changed as follows:

$$\mathcal{C}_{\text{free}} := \{\boldsymbol{c} \mid \mathcal{W}(\boldsymbol{p}) \cap \mathcal{R} \cap \mathcal{O} = \varnothing\}. \tag{14}$$

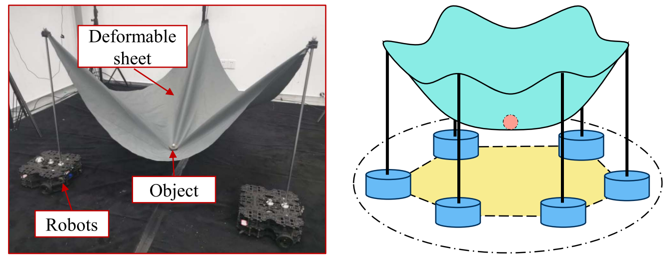

Fig. 7. Multirobot transportation with a deformable sheet as the carrier. (a) Physical model. (b) Simulation model.

The newly added constraints ensure that the object does not collide with robots and obstacles during transportation. Formations that satisfy formation constraints are determined according to (12)–(14).

A. Transportation Without Carrier

In some systems, there are no special carriers, and mobile robots themselves are regarded as carriers. These systems are cheap and easy to control. This is the simplest case because robots are often physically attached to objects, which means there is only one formation s. During transportation, relative positions between robots and objects are invariably kept as the initial. Consequently, valid configuration space is reduce to $\mathcal{C}_{\text{free}} \subset \mathbb{R}^3$. In this case, the whole system can be considered as a rigid body to avoid obstacles.

B. Transportation With Deformable Sheet

Multirobot transportation with a deformable sheet as the carrier is shown in Fig. 7. The transported objects are placed on the deformable sheet held by robots. These systems are universal to different shapes of objects and the position of object can be manipulated by adjusting relative positions between robots. Due to the deformable sheet, there are some constraints from the system itself. For example, due to the physical constraint of deformable sheets, the topological order of robots cannot be changed. Therefore, the conditions of formation switch are more strict.

In order to get the formations, which satisfy constraints, the model of these systems has been studied in our previous work [53]. A computational approach based on the virtual variable cables model (VVCM) is presented for calculating (12), which simplifies the system to a robots-cables-payload system. Then, $\mathfrak{F}$ is replaced by VVCM as follows:

$$\boldsymbol{p} = VVCM(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n). \tag{15}$$

Equation (15) assumes that the position of object is at the state of minimum gravitational potential energy solved by convex optimization. It should be pointed out that the minimal system is composed of three robots. For the minimal systems, (12) can be directly calculated by VVCM. For systems with a higher number of robots, there are multiple local static equilibrium states, resulting in multiple solutions. To deal with the multiple solutions, an optimal formation function is introduced.

$$\begin{aligned} J(s) &= J_{\text{trans}} + J_{\text{pass}} + J_{\text{cross}} \\ s^* &= \text{argmin}\, J(s) \end{aligned} \tag{16}$$

where J_{trans} ensures the safety and stability of systems during transportation. Valid configurations are included in J_{pass} and J_{cross}, where the former is to bypass obstacles, the latter is to cross obstacles. More details about the definition and weight of each item can be found in our previous work [53]. According to (15) and (16), formations $\mathcal{S}$ can be determined for the subsequent planning.

VI. Simulations and Discussions

A. Simulation Setup

In this section, we have designed three scenarios for MMRSs under different formation constraints to test our planner. These systems are placed in an environment containing a large number of randomly generated obstacles. All our simulations are performed on a computer with Intel Core i7-10700 CPU at 2.90 GHz (16 cores) and 64 GB RAM. The results are obtained by the simulation platforms ROS-Gazebo and ROS-Rviz.

B. Simulation Results

1) Maintaining Rigid Formations: In the first part, mobile robots are required to form a rigid formation while traversing an environment with obstacles. Since the formation is rigid, which indicates that $\mathcal{S}$ is onefold, the valid configuration space is reduced to 3-D. We set up an environment of 10 m $\times$ 10 m. The initial position of formation center is (1.5, 1.5) m, and the target is (8.5, 8.5) m. On the premise of ensuring that the initial and target position exists, some obstacles with random positions are generated in the environment, whose radiuses are also randomly selected between 0.05 m and 0.1 m. The radius of each robot is 0.35 m. The results are shown in Fig. 8. In Fig. 8(a), four robots try to maintain a square formation. There are 65 obstacles in the environment. The average path length of each robot is 19.1599 m on the optimally planned path. Similarly, in Fig. 8(b), six robots try to maintain a hexagon, and there are 90 obstacles in an environment of 15 m $\times$ 15 m. The initial position of formation center is (2.5, 2.5) m while the target is (12.5, 12.5) m. The average path length of each robot is 24.7397 m. It can be seen that the two MMRSs are able to traverse the obstacle-rich environments while holding desired formations invariably unchanged. To our best knowledge, no existing planner for multirobot systems can deal with these environments.

It should be pointed out that our planner may also fail in these environments since all obstacles are randomly generated. Therefore, we repeat lots of simulations for the system composed of four robots under different numbers of obstacles. Moreover, in order to study the effect of discrete scales in the planner, we also repeat under different scales. The results are given in Table II. For each time, obstacles are randomly generated while the initial and goal positions are fixed, and the environment is discretized

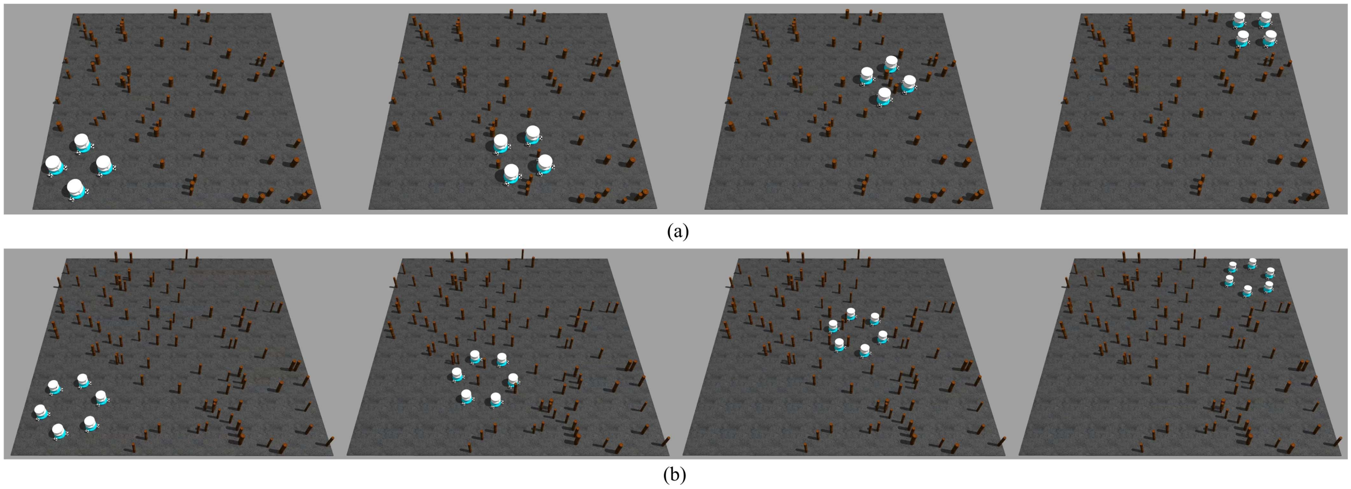

Fig. 8. MMRSs traverse obstacle-rich environments while maintaining rigid formations. (a) Four robots form a square in a 10 m × 10 m environment. There are 65 obstacles with random positions and radii. (b) Six robots form a hexagon in a 15 m × 15 m environment. There are 90 obstacles with random positions and radii. (a) A four-robot system. (b) A six-robot system.

TABLE II
REPEAT FIG. 8(a) UNDER DIFFERENT CONDITIONS WITHOUT BOUNDARY DENSIFICATION

Obstacles	Discrete Scale: g = 0.08m α = 0.04 rd				Discrete Scale: g = 0.05m α = 0.04 rd				Discrete Scale: g = 0.025m α = 0.04 rd			
	Success Rate	Average Time (s) Mapping	Planning	Total	Success Rate	Average Time (s) Mapping	Planning	Total	Success Rate	Average Time (s) Mapping	Planning	Total
10	100/100	2.12479	0.50741	2.63220	100/100	5.26545	1.39201	6.65746	100/100	20.87622	5.98285	26.85907
20	99/100	2.18363	0.34327	2.52690	99/100	5.42517	0.82054	6.24571	99/100	21.50460	3.87619	25.38079
30	93/100	2.22332	0.22335	2.44667	**93/100**	5.53143	0.59595	**6.12738**	**94/100**	21.93061	2.76713	**24.69774**
40	**67/100**	2.25200	0.14657	**2.39857**	**70/100**	5.59970	0.40005	**5.99975**	**77/100**	22.30289	1.80580	**24.10869**
50	**36/100**	2.28206	0.10722	**2.38928**	**41/100**	5.67768	0.31582	**5.99350**	**50/100**	22.64238	1.31906	**23.96144**
60	23/100	2.31072	0.07745	2.38817	**23/100**	5.78103	0.21294	**5.99397**	**26/100**	22.95013	0.91771	**23.86784**
70	2/100	2.32667	0.05235	2.37902	**2/100**	5.84270	0.15498	**5.99768**	**5/100**	23.17927	0.66619	**23.84546**
80	0/100	2.34902	0.04201	2.39103	1/100	5.89965	0.11418	6.01383	1/100	23.34135	0.50529	23.84664
90	1/100	2.36974	0.03097	2.40071	1/100	5.95067	0.10586	6.05653	1/100	23.69105	0.42129	24.11234
100	0/100	2.38406	0.02386	2.40792	0/100	5.98362	0.07432	6.05794	0/100	23.90271	0.31573	24.21844

TABLE III
REPEAT FIG. 8(a) UNDER DIFFERENT CONDITIONS WITH BOUNDARY DENSIFICATION

Obstacles	Discrete Scale: g = 0.08m/0.04m α = 0.04 rd				Discrete Scale: g = 0.05m/0.025m α = 0.04 rd				Discrete Scale: g = 0.025m/0.0125m α = 0.04 rd			
	Success Rate	Average Time (s) Mapping	Planning	Total	Success Rate	Average Time (s) Mapping	Planning	Total	Success Rate	Average Time (s) Mapping	Planning	Total
10	100/100	2.73023	1.19019	3.92042	100/100	6.57826	2.66315	9.24141	100/100	26.94437	10.36484	37.30921
20	99/100	2.77770	0.91127	3.68897	99/100	6.50419	1.97885	8.48304	99/100	26.21870	7.43409	33.65279
30	93/100	2.80989	0.74911	3.55900	**94/100**	6.42345	1.46426	**7.88771**	94/100	25.83864	5.50717	31.34581
40	**69/100**	2.76320	0.53806	**3.30126**	**72/100**	6.38639	1.06474	**7.45113**	**80/100**	26.55954	4.30486	**30.86440**
50	**37/100**	2.75822	0.40120	**3.15942**	**47/100**	6.48182	0.84711	**7.32893**	**52/100**	25.91642	3.25864	**29.17506**
60	23/100	2.67300	0.29691	2.96991	**24/100**	6.47801	0.64271	**7.12072**	**27/100**	25.79795	2.43327	**28.23122**
70	2/100	2.65702	0.22995	2.88697	**3/100**	6.33030	0.45257	**6.78287**	**6/100**	25.91122	1.93100	**27.84222**
80	0/100	2.62448	0.17087	2.79535	1/100	6.28102	0.34993	6.63095	1/100	24.95567	1.43521	26.39088
90	1/100	2.58503	0.12706	2.71209	1/100	6.33555	0.29529	6.63084	1/100	26.19357	1.29228	27.48585
100	0/100	2.50583	0.07612	2.58195	0/100	6.25361	0.22073	6.47434	0/100	25.32007	0.93960	26.25967

only once by given scales. It can be seen that under the same discrete scales, as the number of obstacles increases, the success rate generally decreases, and the time spent on mapping from workspace to configuration space increases while the time spent on planning on the connected graph decreases. Essentially, the whole configuration space is limited and the same for determined discrete scales. When obstacles increase, the time for validating valid configurations increases, and obviously, the size of valid configuration space decreases. Then, the size of connected graph decreases, resulting in less time for planning. It can also be seen that for the same number of obstacles, the success rate and total time are both larger under smaller discrete scales.

Smaller discrete scales can increase the success rate of planning but with significant additional computational costs. Thus, we try our proposed boundary densification method, and the results are given in Table III. Compared to the discrete scales $g = 0.05$ m in Table II, boundary densification $g = 0.05/0.025$ m will impose additional computational costs. Correspondingly, the success rate is also improved, as the bolded entries show. However, in some scenarios, the success rate remains the

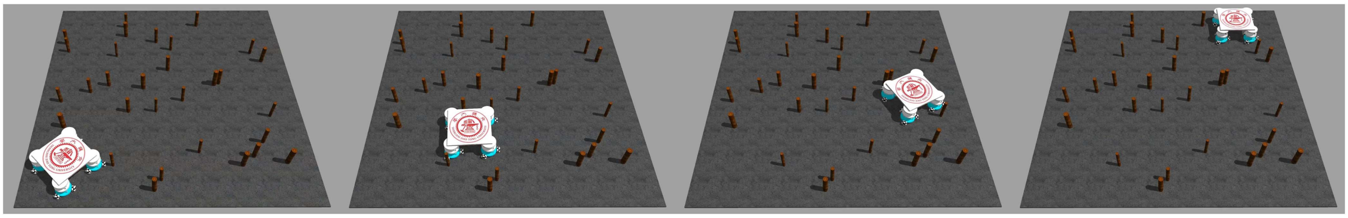

Fig. 9. MMRS composed of four robots transports a square object in a 10 m × 10 m environment. There are 30 obstacles with random positions and radii. The system is forced to bypass obstacles like a 2-D whole.

same after applying our boundary densification method, which is mainly attributed to two reasons. First, the obstacles are generated randomly, and the number of repeated simulations may not be sufficient. Second, the environments have contained too dense obstacles for it to plan paths. It can be seen that when the number of obstacles exceeds 80, it becomes almost impossible to find paths, regardless of smaller discrete scales or boundary densification method. In summary, boundary densification can enhance the success rate of planning to a certain degree.

Simulation results indicate that to increase the success rate, smaller discrete scales or boundary densification method can be used. Comparing these two strategies, it can be found that the additional computational costs from boundary densification are much lower than the smaller discrete scales.

2) Object Transportation Without Carrier: In the second part, mobile robots are required to transport objects without a carrier, which also needs robots to maintain rigid formations. But different from the first part, there are additional constraints brought by objects. Robots and environment are the same as Fig. 8(a). The transported object is a square with a side length of 1.5 m. There are 30 obstacles randomly generated, and the results are shown in Fig. 9. The system also can reach the goal position without collision, and the average path length of each robot is 18.4439 m. Since obstacles are all higher than the system, constraints of formations and safe transportation can be synthesized, which is that the 2-D outline of the whole system does not overlap obstacles. Compared with the first part, the obstacle-crossing ability of this system is observably reduced. It has to merely bypass obstacles and cannot cross them.

3) Object Transportation With Deformable Sheet: In the third part, mobile robots are required to transport objects with a deformable sheet, the model of which has been studied in our previous work [53]. Now, formation constraints are no longer rigid, and the system is able to switch formations to bypass or cross obstacles depending on the height of transported object. We first test a three-robot system. Each side length of the original deformable sheet is 1.6 m, and the height of each contact point between robots and the sheet is 1 m. These parameters will be used to calculate the model as described in (15). The environment is also 10 m × 10 m, and the radius of each robot is 0.15 m. The initial position of formation center is (1.5, 1.5) m, and the target is (8.5, 8.5) m. There are 80 obstacles randomly generated with random radius between 0.1 m and 0.15 m, among which 30 are 0.25 m in height, 40 are 0.42 m in height, and 10 are 1 m in height. Limited by formation constraints and the deformable sheet, the system has to bypass some obstacles. λ_s is set as the smallest for the formation of a triangle with each side length of 0.7 m, which keeps the object at 0.35 m in height. The system will try to maintain this formation and formation switch only occurs whenever necessary. The results are shown in Fig. 10, and the deformable sheet is simplified to a robots–cables–payload system as described before. Obstacles of different heights are distinguished by different colors. The system can traverse this obstacle-rich environment without collision. The average path length of each robot is 17.1758 m. In Fig. 10(c), it can be seen that the system has to switch formations to uplift transported object to cross some obstacles. The height of object during transportation is shown in Fig. 10(d).

C. Comparison With Other Methods

We then test a four-robot system in a more complex environment including a narrow corridor. The environment is 16 m × 4 m and contains 70 obstacles. λ_s is set as the smallest for the formation of a square with each side length of 1.1314 m. The results are shown in Fig. 11.

As introduced before, there are three categories of approaches. For comparison, we tried the first category approaches [10], [11], [12], which consist of two behaviors: leader–follower for forming formations and artificial potential field for avoiding obstacles. At first, this approach was run in the same environment as Fig. 11. However, robots always collided with obstacles regardless of how the weights between the two behaviors are adjusted. Therefore, we decided to test the approach in a simpler environment as shown in Fig. 12, where there are 16 obstacles and robots need to maintain a regular triangle. It can be seen rigid formation can be held when robots are not affected by obstacles. When crossing obstacles, deformations will appear as shown in Fig. 12(b). Moreover, the quantifying relationship between degree of deformations and weights coupling two behaviors is not clear, resulting in difficulty in controlling formations. This method is not appropriate in such obstacle-rich environments.

We also tried the second category method as in [22]. The strategy is to find the largest obstacle-free space containing the system, inside which motions of robots can be planned. The result is shown in Fig. 13. The red polygon is the largest obstacle-free space currently containing the robot system. Motions of robots can be planned freely in the red polygon which changes with the movement of system. It can greatly avoid collisions and maintain desired formations. However, the method failed after several movements because the obstacle-free space could not

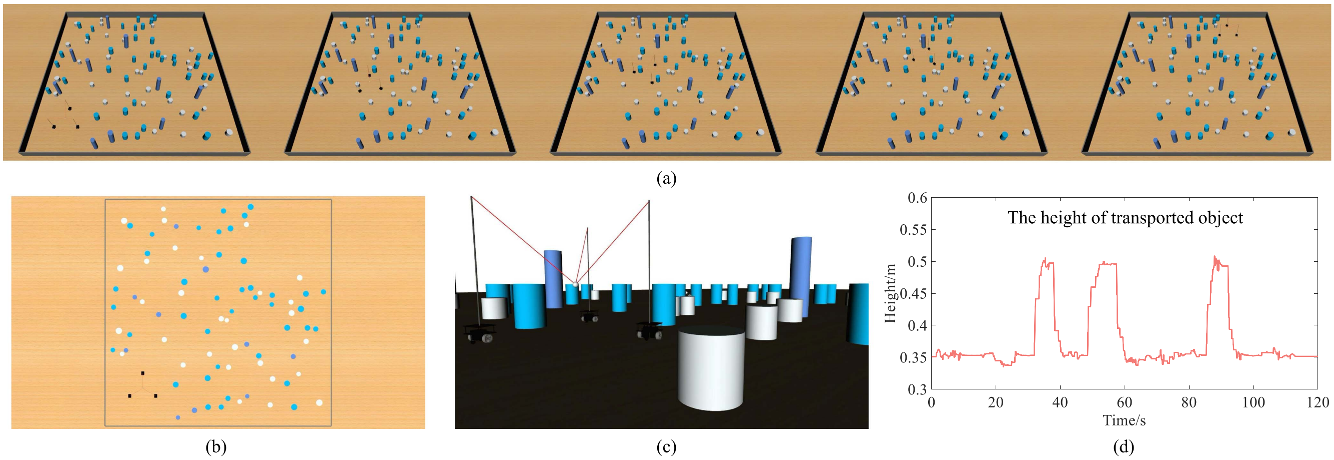

Fig. 10. MMRS composed of three robots transport objects with a deformable sheet in a 10 m × 10 m environment. The deformable sheet is simplified to a robots-cables-payload system. There are 80 obstacles with random positions and radii. Obstacles of different heights (0.25 m, 0.42 m, 1 m) are distinguished by different colors. (a) Pictures during transportation. (b) Top view of the whole environment. (c) Robot's view. It can be seen that some obstacles can only be bypassed. (d) The height of object during transportation.

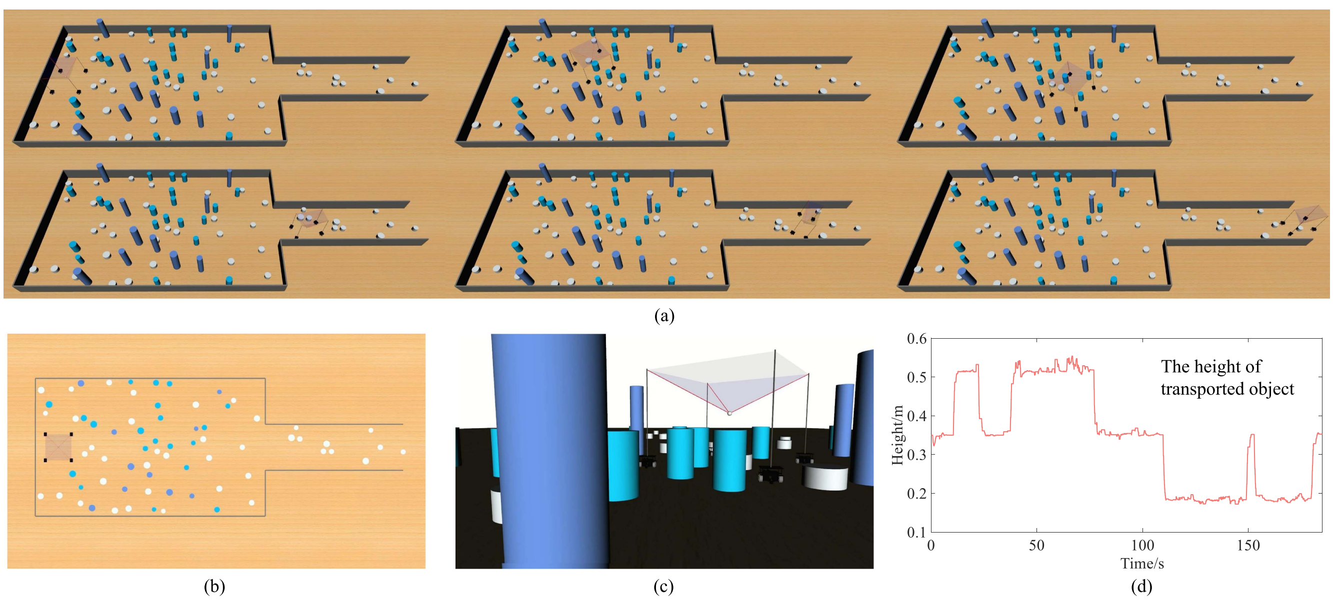

Fig. 11. MMRS composed of four robots transports objects with a deformable sheet in a 16 m × 6 m environment. There are 70 obstacles. (a) Pictures during transportation. (b) Top view of the whole environment. (c) Robot's view. (d) Height of object during transportation.

be found anymore for the system to move toward the destination. This method is also not appropriate in such obstacle-rich environments.

In contrast to methods of the two categories, our planner can simultaneously satisfy formation and obstacle constraints by planning the path in valid configuration space. As for the difference between our planner and some other graph-based planner, it is in the different uses of the graph. In our planner, a vertex on the graph represents a valid configuration instead of a robot's position, while edges represent the connectivity between valid configurations. In this regard, the valid configuration space is represented by a graph. Accordingly, efficient graph search methods can be conveniently employed, and optimal paths can also be directly searched. Furthermore, the cost functions can be designed based on different requirements of formation preference.

D. Discussion on Proposed Method

In our planner, the valid configuration space of a MMRS is built, in which both formation and obstacle constraints can be satisfied. Then, graphs are used to describe the valid configurations and costs between them. Besides, a boundary densification method is proposed to further improve the search performance. The planner can deal with different formation constraints and additional constraints brought by special tasks. We take object transportation as a typical example to test our planner.

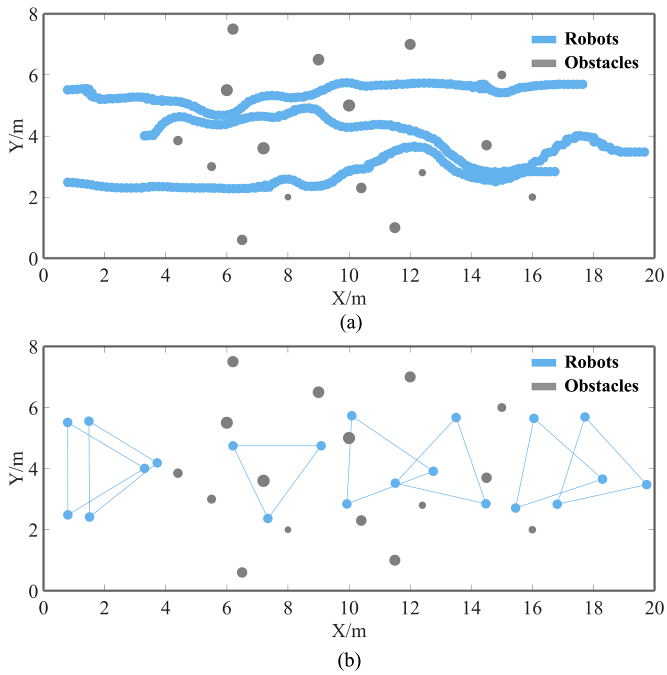

Fig. 12. Motion planning by the category of method in [10], [11], [12], which consists of avoiding obstacles and forming formations. (a) Paths of robots. (b) Deformations of formation during movement.

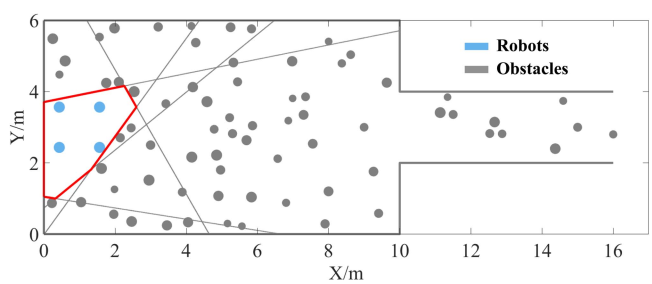

Fig. 13. Motion planning by the method in [22]. The largest obstacle-free space currently containing the robot system is denoted by the red polygon. Robots can move freely in the red polygon. However, it cannot drive the system to the destination if considering the formation constraints.

Simulation results show that the planner can be applied to obstacle-rich environments.

However, our planner also has some limitations. First, parameters, which dominate the performance of our planner, are usually not universal in different environments. The planner will fail if the discrete scales are set inappropriately. Although we have demonstrated the impact of different discrete scales through a large number of simulations, it is still necessary for the user to make specific choices according to different situations. Secondly, the planner relies on a priori global information about the environment and cannot be applied in environments with dynamic obstacles. Before searching paths, the valid configuration space of the system is required to be built and mapped to the graph, which is related to the workspace. Thus, if the workspace changes, the valid configuration space needs to be rebuilt. Avoiding dynamic obstacles is not possible currently since the mapping time is too heavy for real-time.

The simulation results have proven the effectiveness of our planner. Still, there are several technical challenges in the implementation of our planner on actual hardware. First, it is not easy to obtain accurate absolute positions of robots and obstacles in the real world. This usually requires the use of an external wide-area positioning system, which is further desired with a high degree of accuracy. Second, during movement between planned path points, it is difficult to maintain formations since synchronization between different robots is a big problem in hardware. Moreover, different kinematic models of robots such as nonholonomic robots will also lead to difficulty in maintaining formations. Finally, stable control laws are needed to eliminate the errors resulting from executing the planned motions.

VII. Conclusion

This article proposes a novel multimobile robot motion planner with excellent obstacle avoidance ability under formation constraints. The optimal motions of robots can be planned in obstacle-rich environments. Valid configurations of systems are defined to satisfy both formation and obstacle constraints. Then, the valid configuration space is built and mapped to an undirected graph. Edges of the graph are checked based on the connectivity between configurations. Then, efficient graph search algorithms can be directly used. Moreover, a boundary densification method is used for improving the success rate of planning. The generality and effectiveness of the planner are proved by implementing different cases. Despite taking specific examples, the framework of our planner can be potentially employed in other scenarios.

Regarding future work, we will further study the proposed method in environments with dynamic obstacles. It may be unnecessary to recreate the complete valid configuration space since the dynamic environment will only bring a small percentage of changes. In this regard, a possible strategy is to combine the planner as a global planner with some local obstacle avoidance methods. When the MMRS approaches a dynamic obstacle, local methods will be employed for real-time obstacle avoidance. More practically, we will try to carry out experiments on actual hardware. By dealing with different kinematic models of robots, we can further refine our planner. If possible, we will also integrate a suitable controller to make our approach more complete.

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